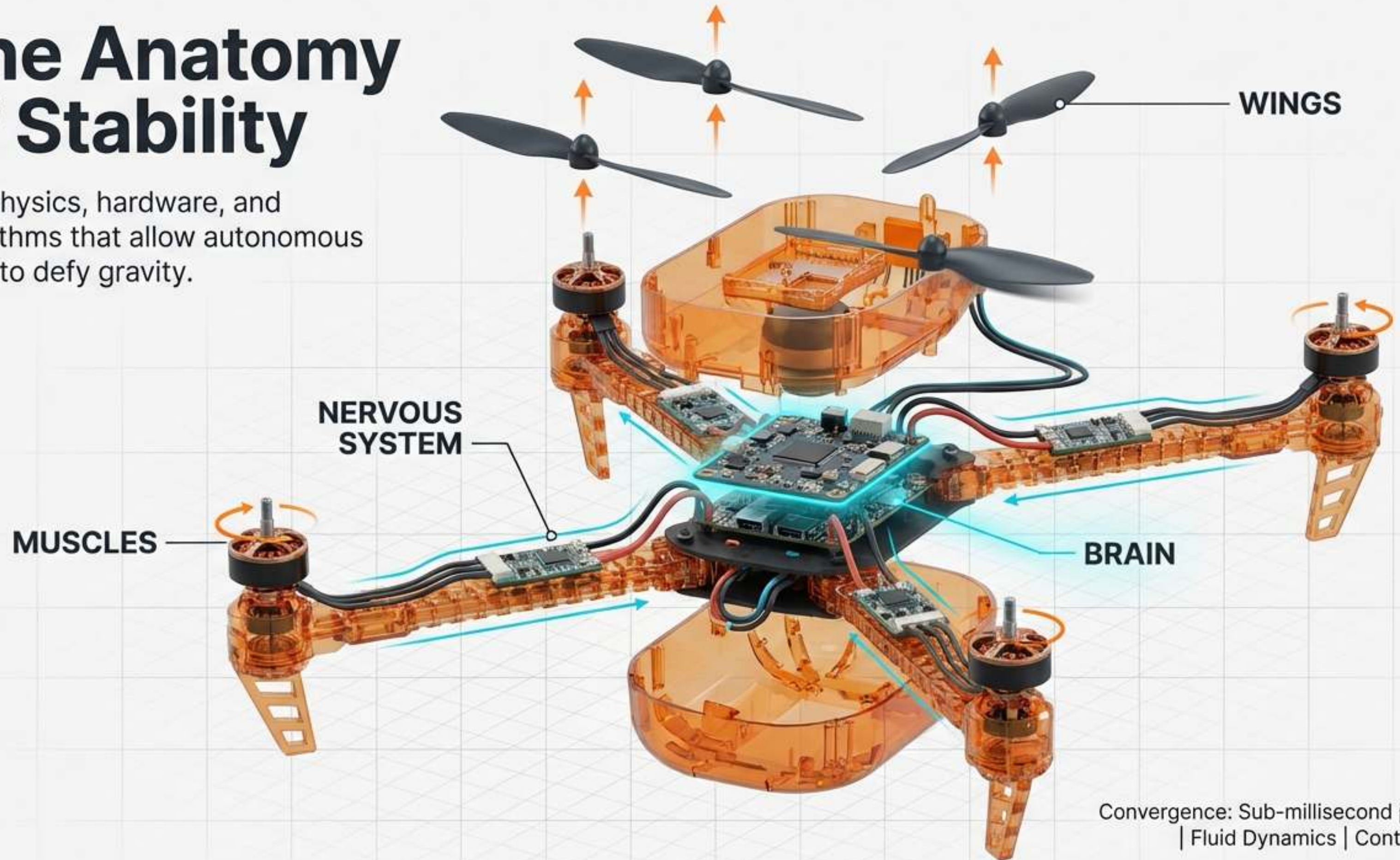


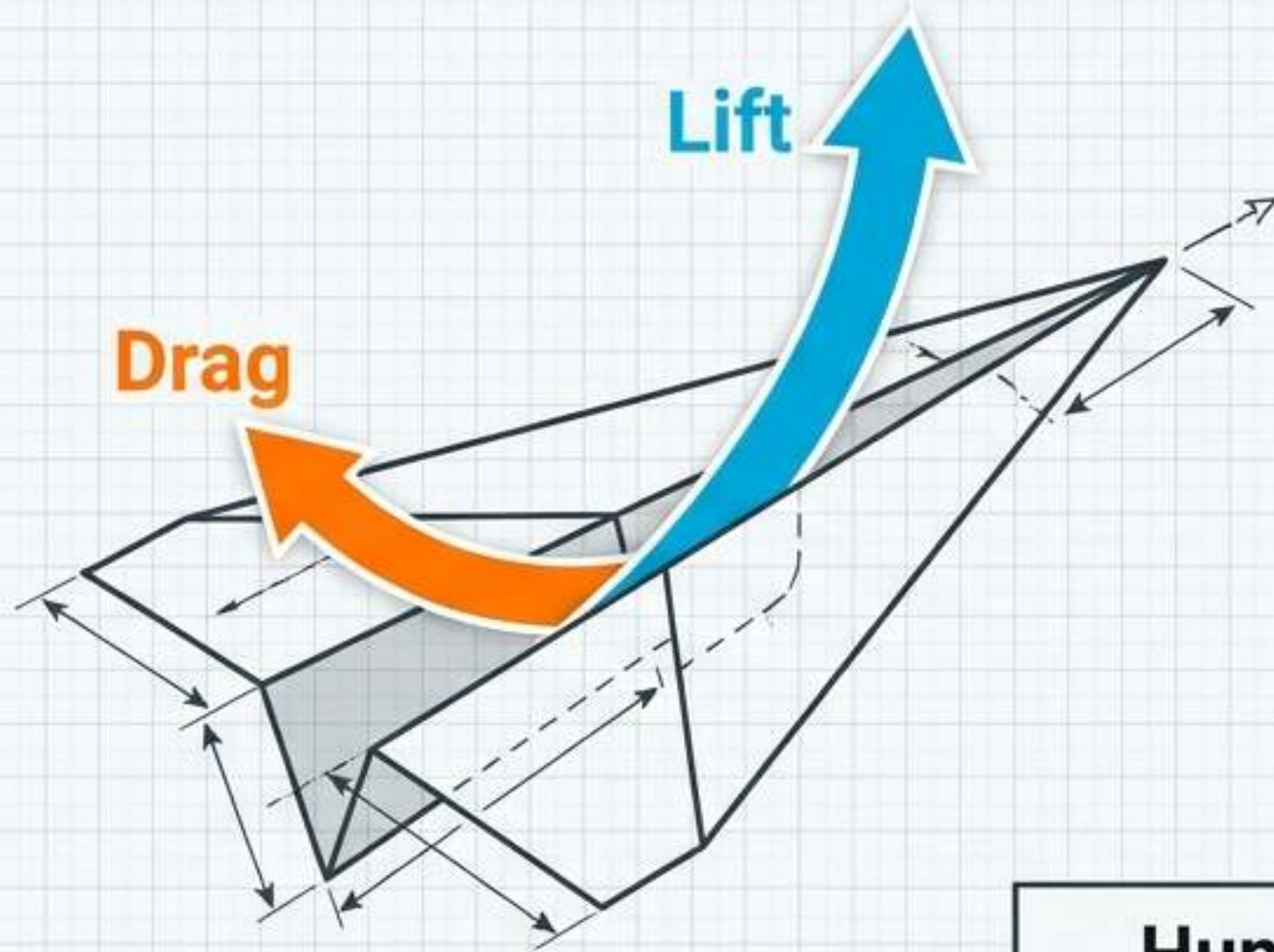
The Anatomy of Stability

The physics, hardware, and algorithms that allow autonomous flight to defy gravity.



Convergence: Sub-millisecond processing
| Fluid Dynamics | Control Theory.

NATURAL STABILITY

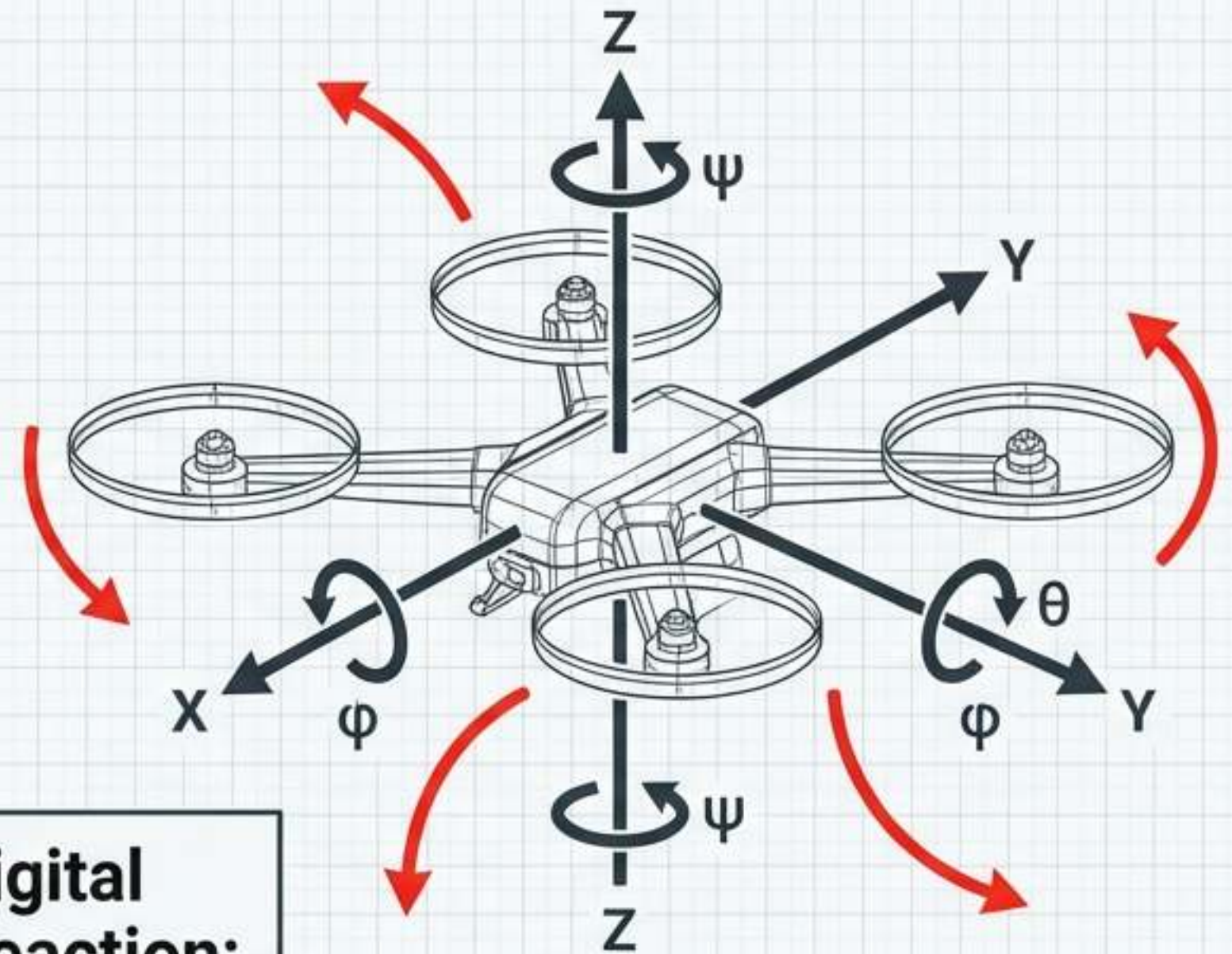


Aerodynamic forces passively balance to maintain flight path.

Human Reaction:
~200ms

Digital Reaction:
<1ms

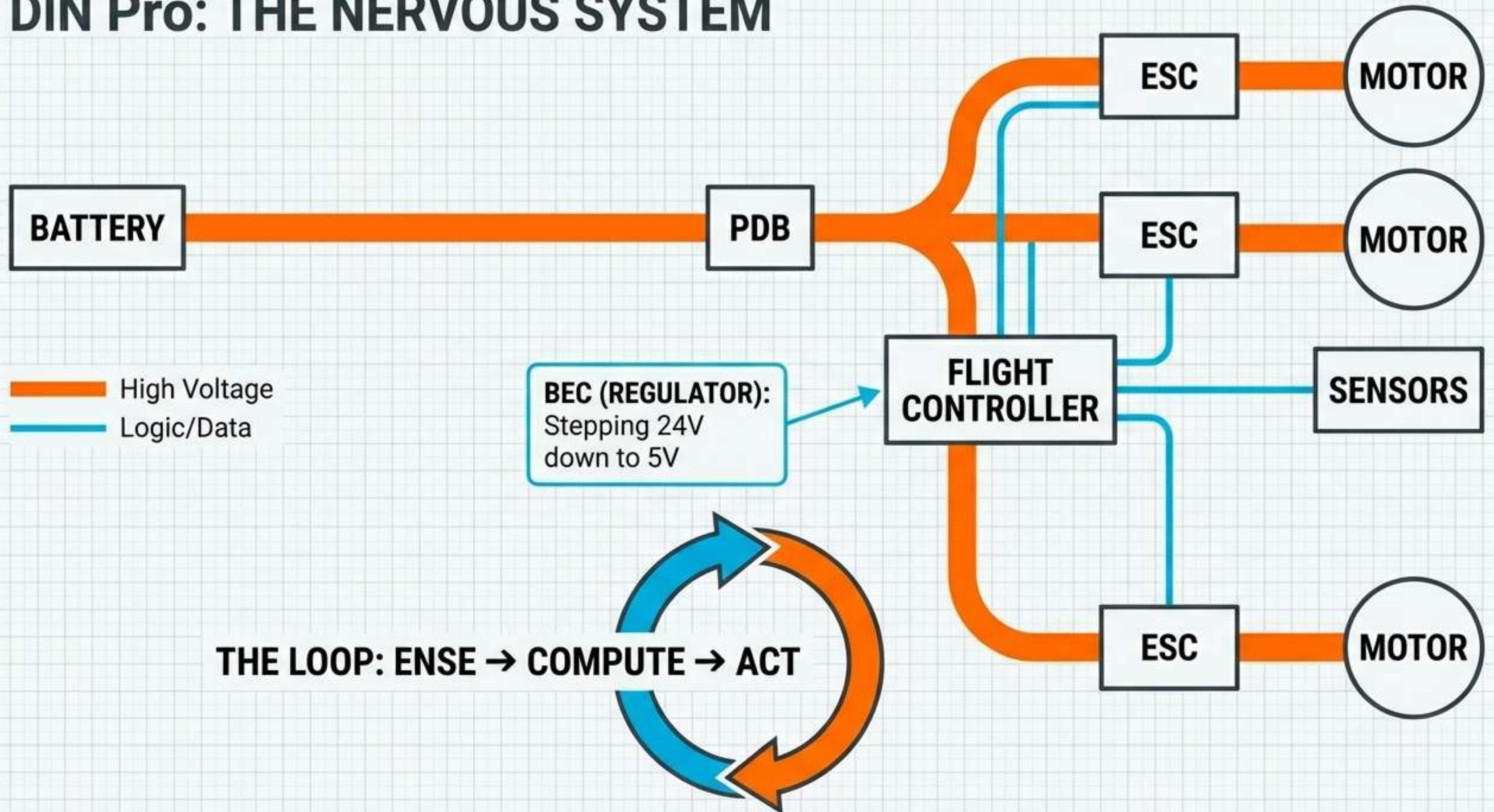
INHERENT INSTABILITY



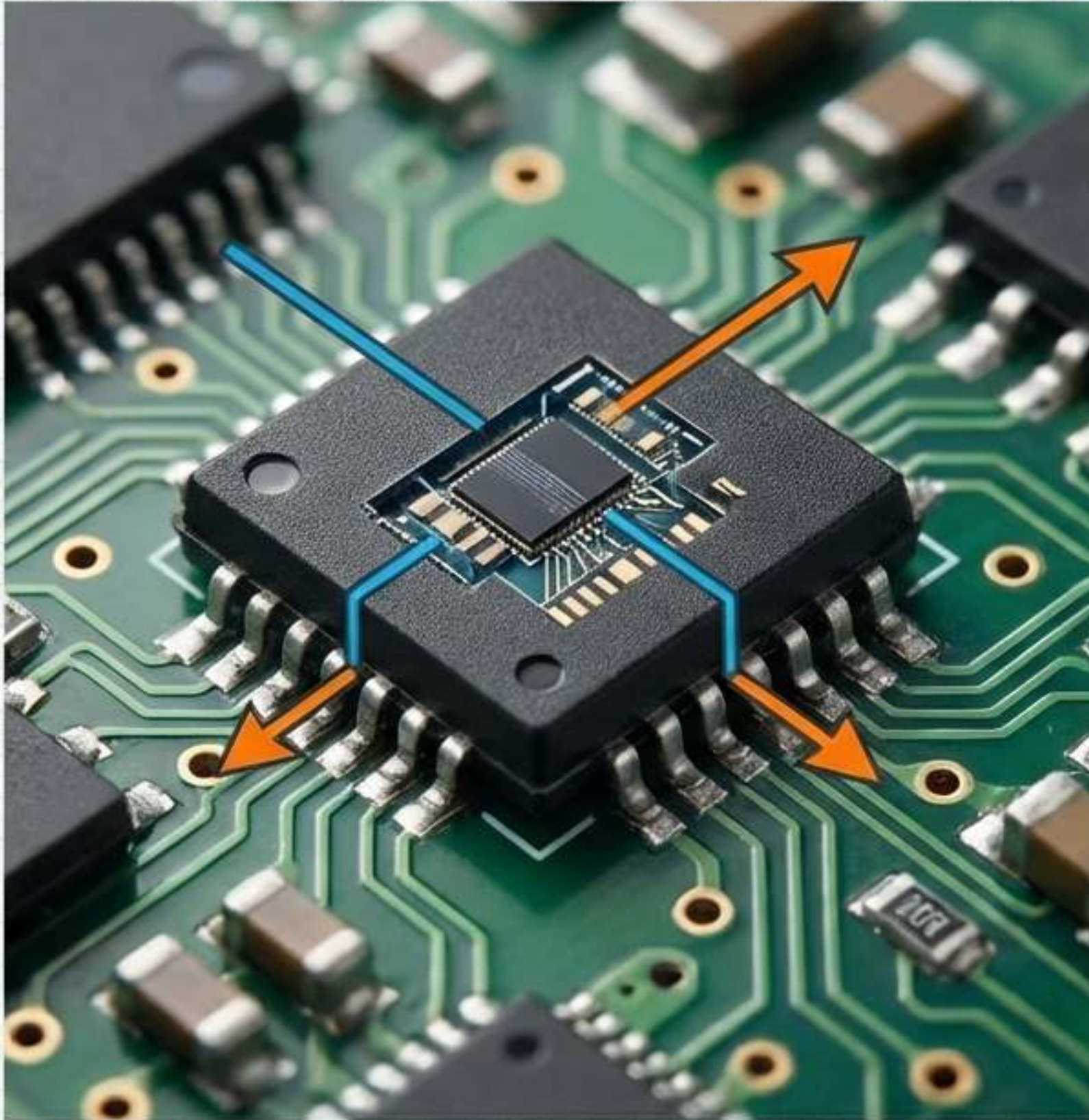
Requires continuous active corrections across all 6 axes.

Unlike a plane, a drone is a brick fighting gravity.
Without active intervention, it crashes in seconds.

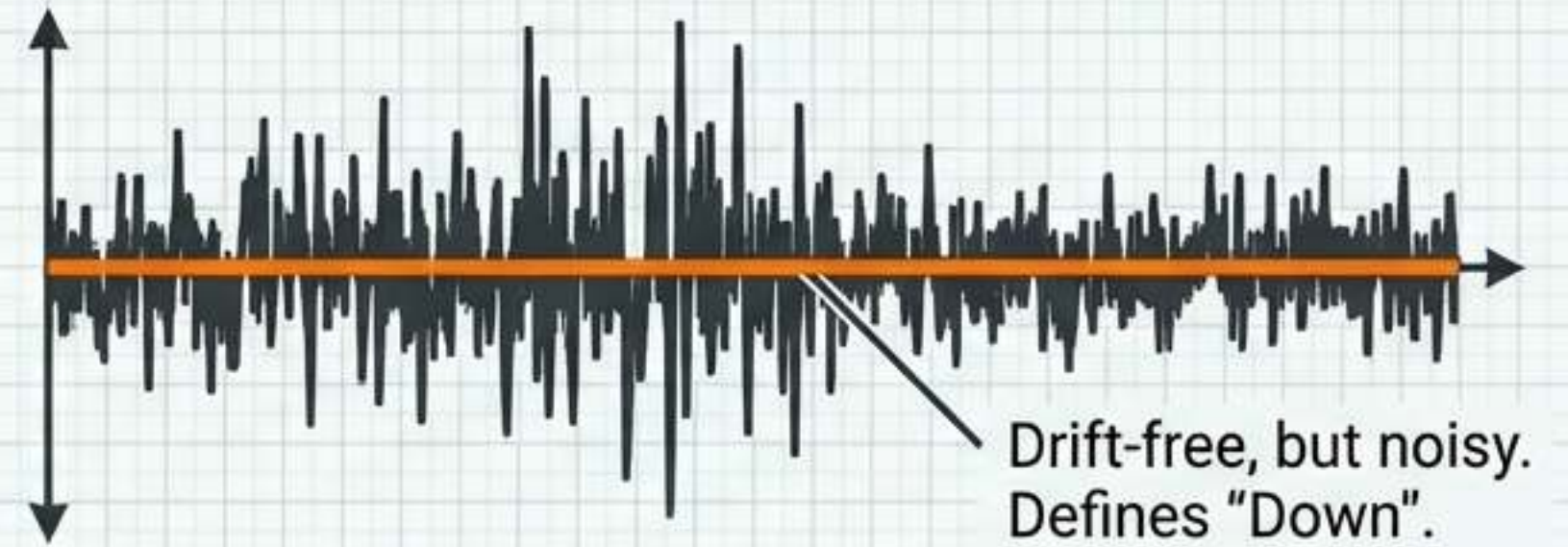
DIN Pro: THE NERVOUS SYSTEM



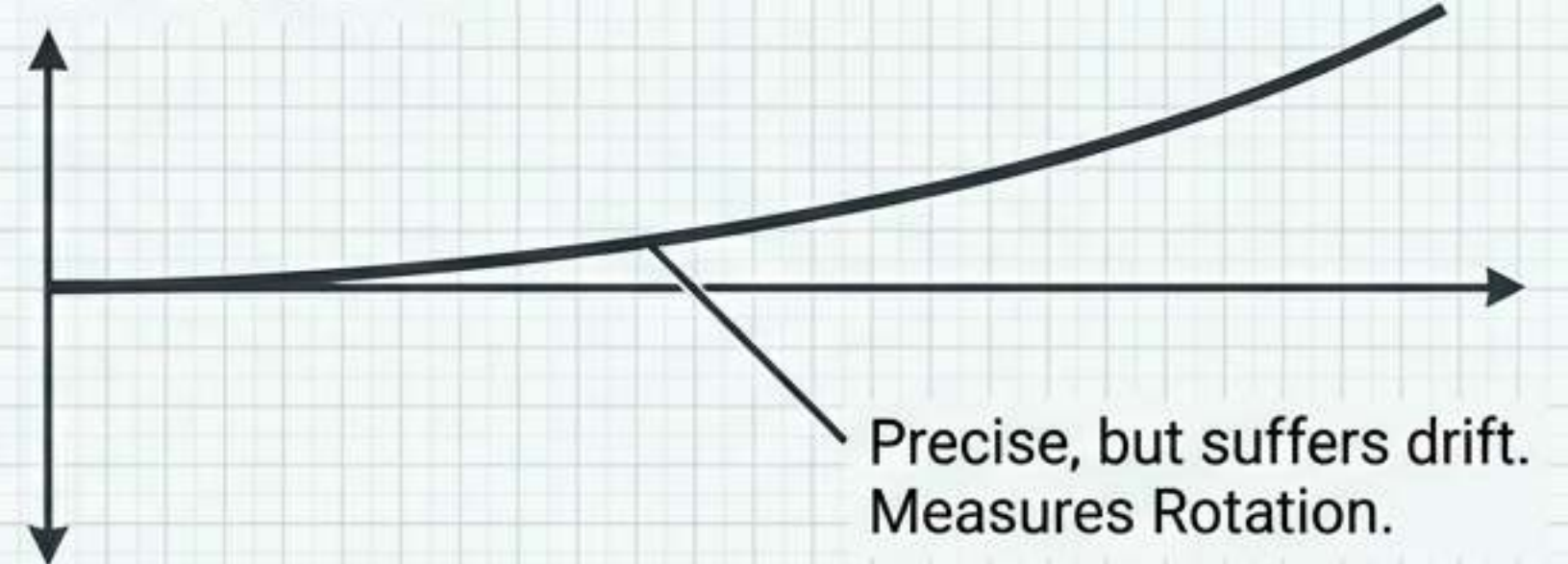
THE INNER EAR: THE IMU



ACCELEROMETER



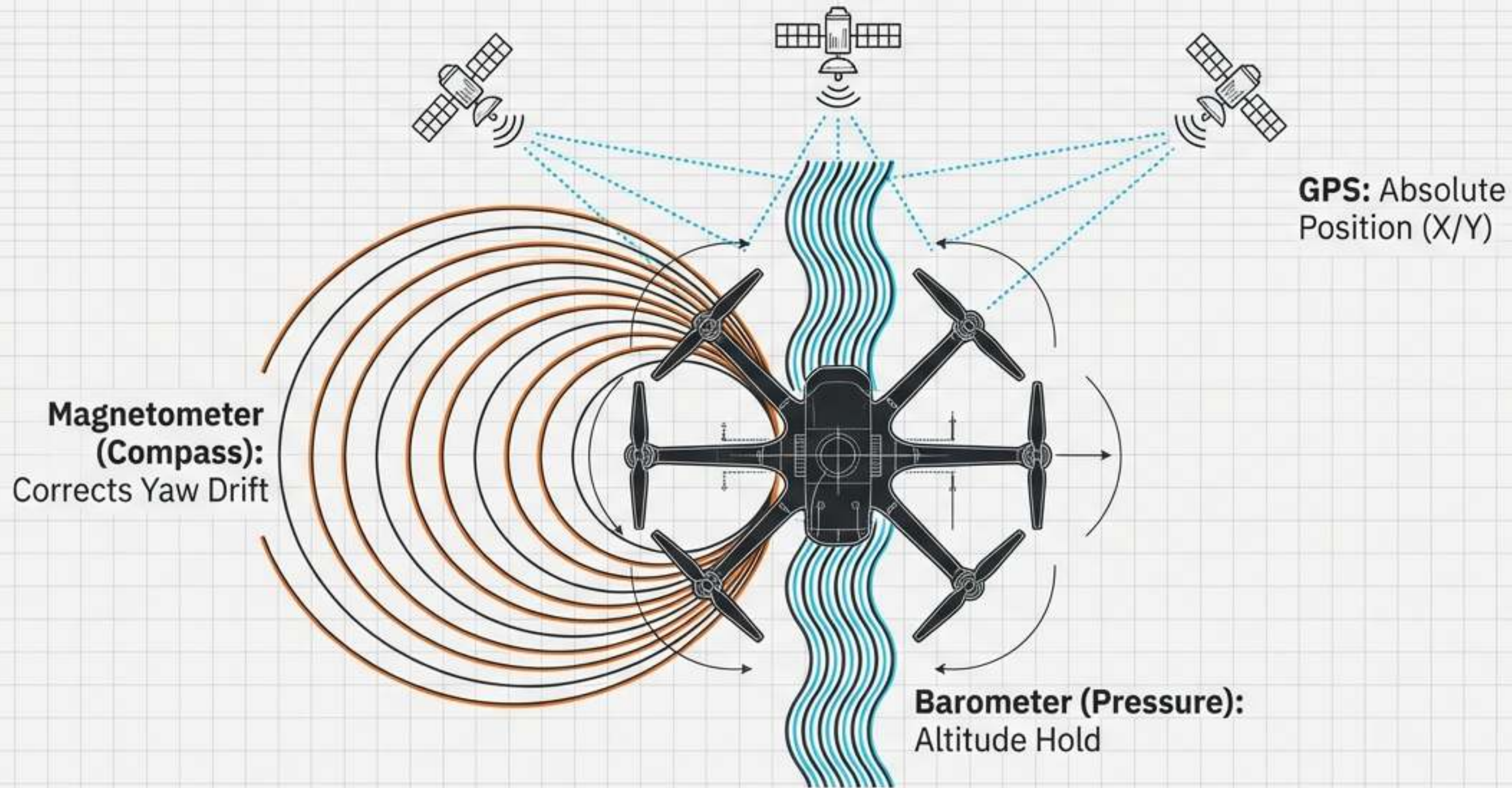
GYROSCOPE



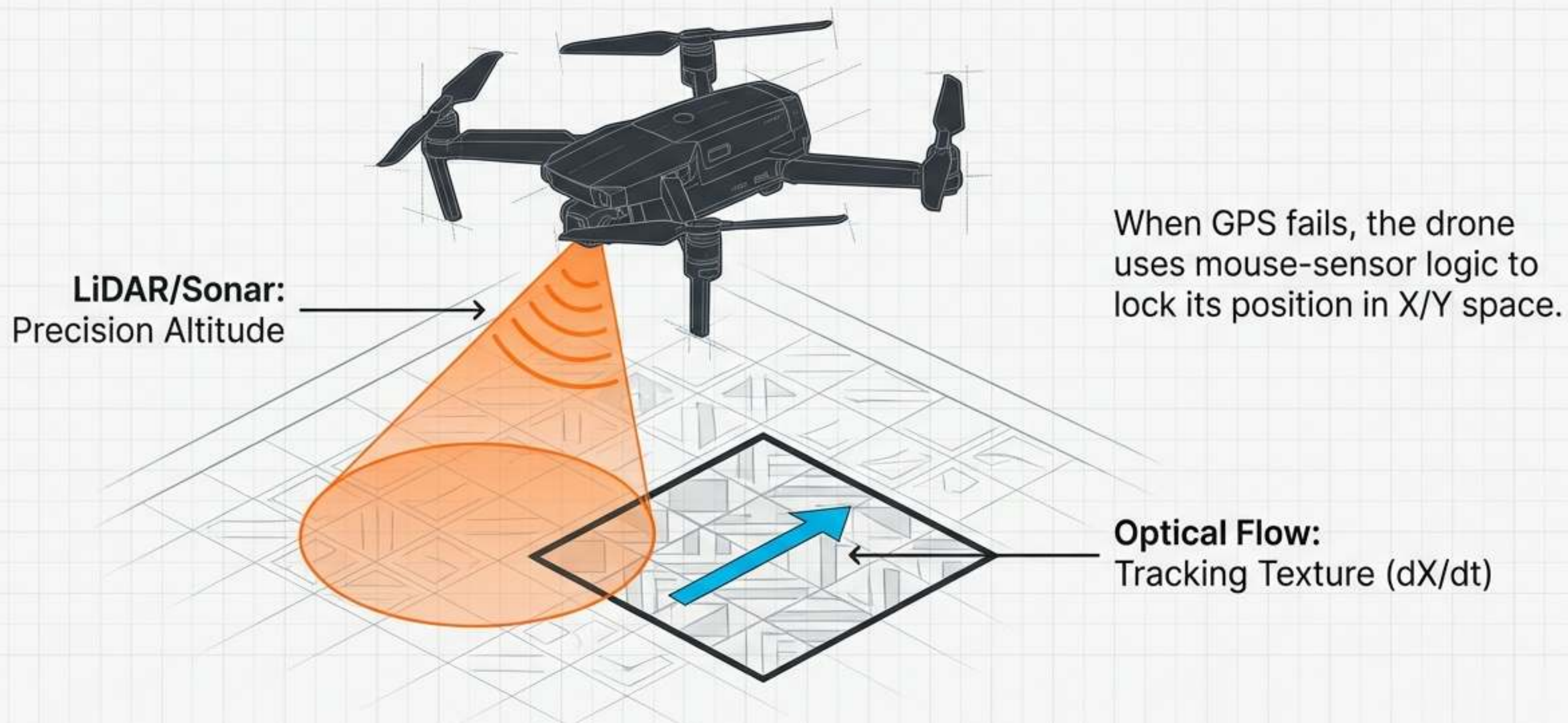
The IMU combines these to know 'Down' and 'Spin'.

Spatial Awareness

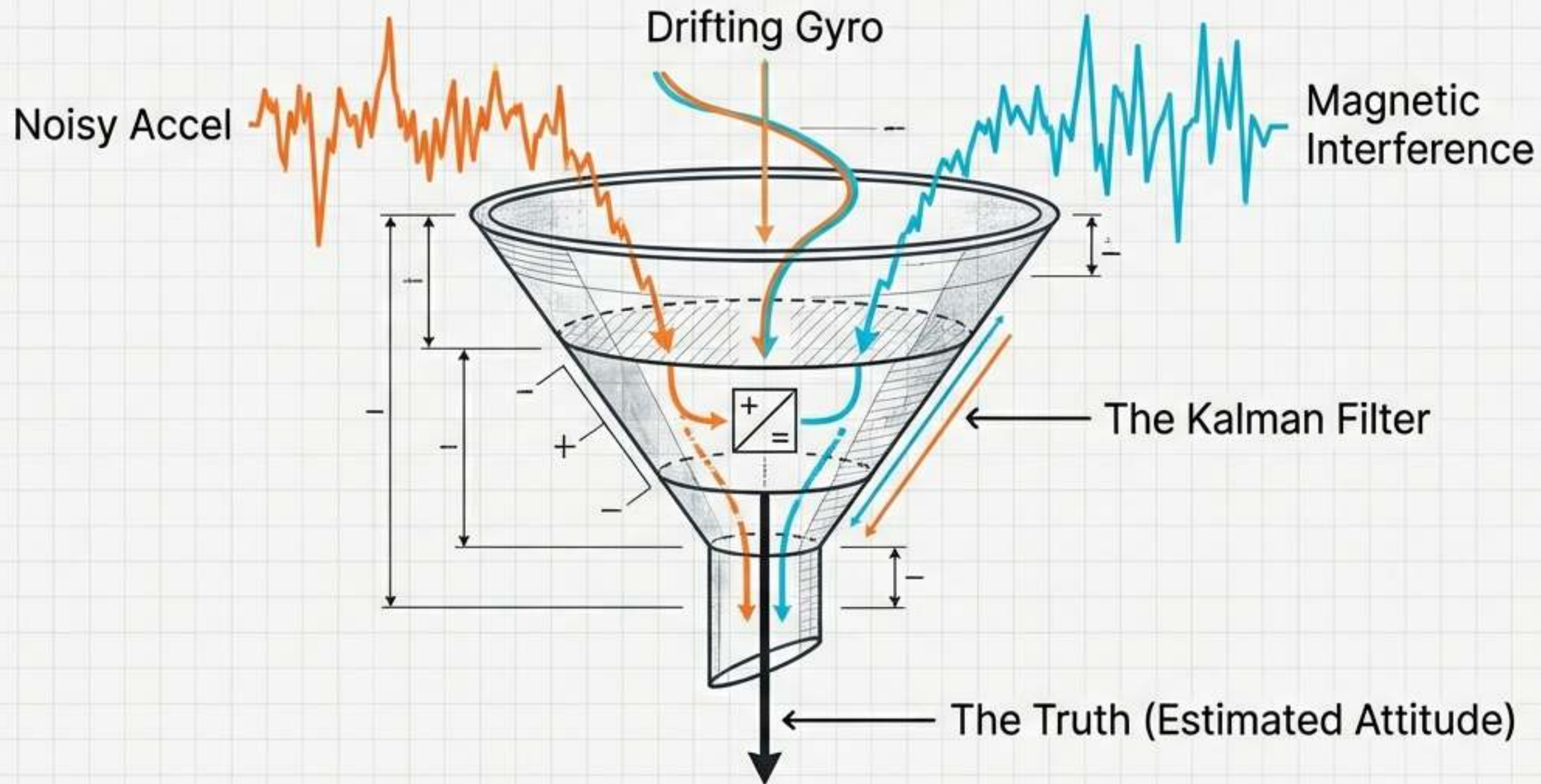
Placing the organism in the physical world



SIGHT BEYOND SATELLITES

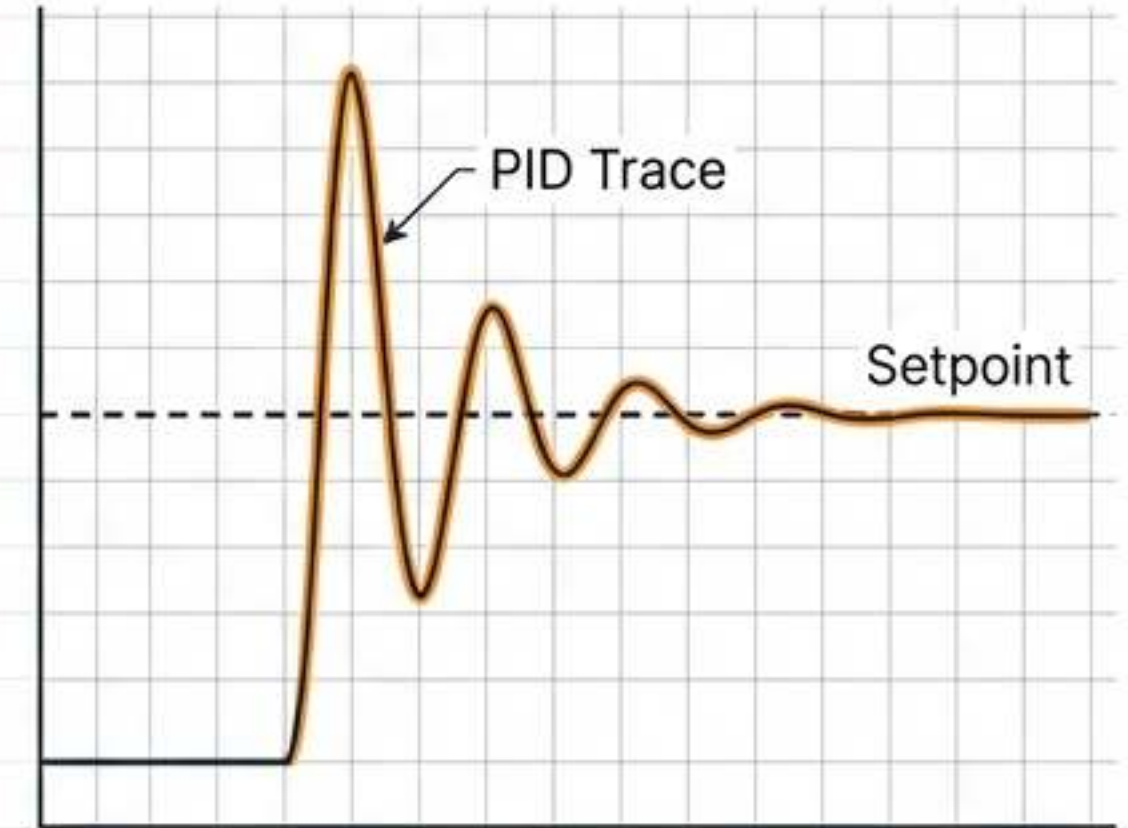
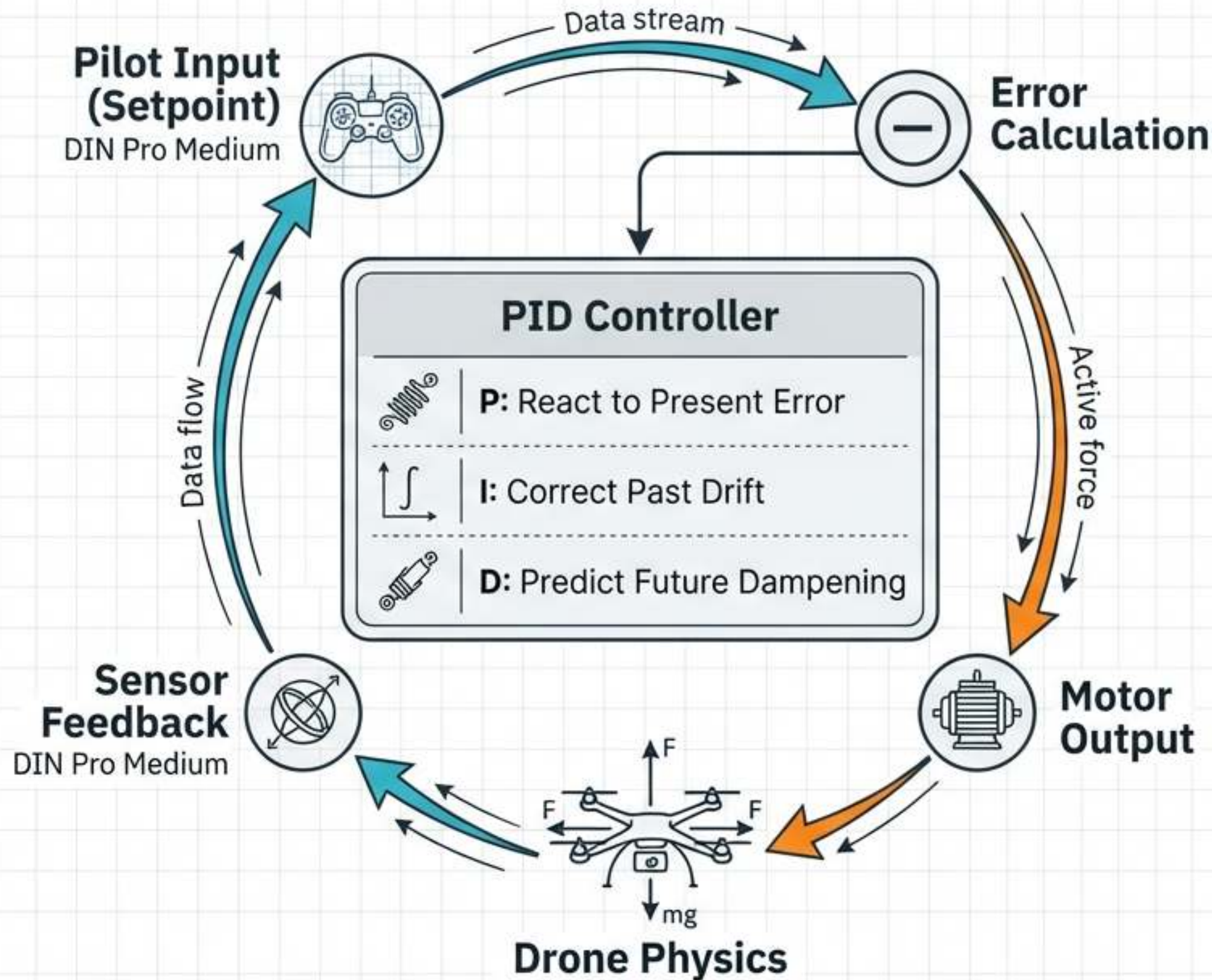


The Brain I: Sensor Fusion



Sensors lie. The brain must mathematically fuse them to find the true orientation.

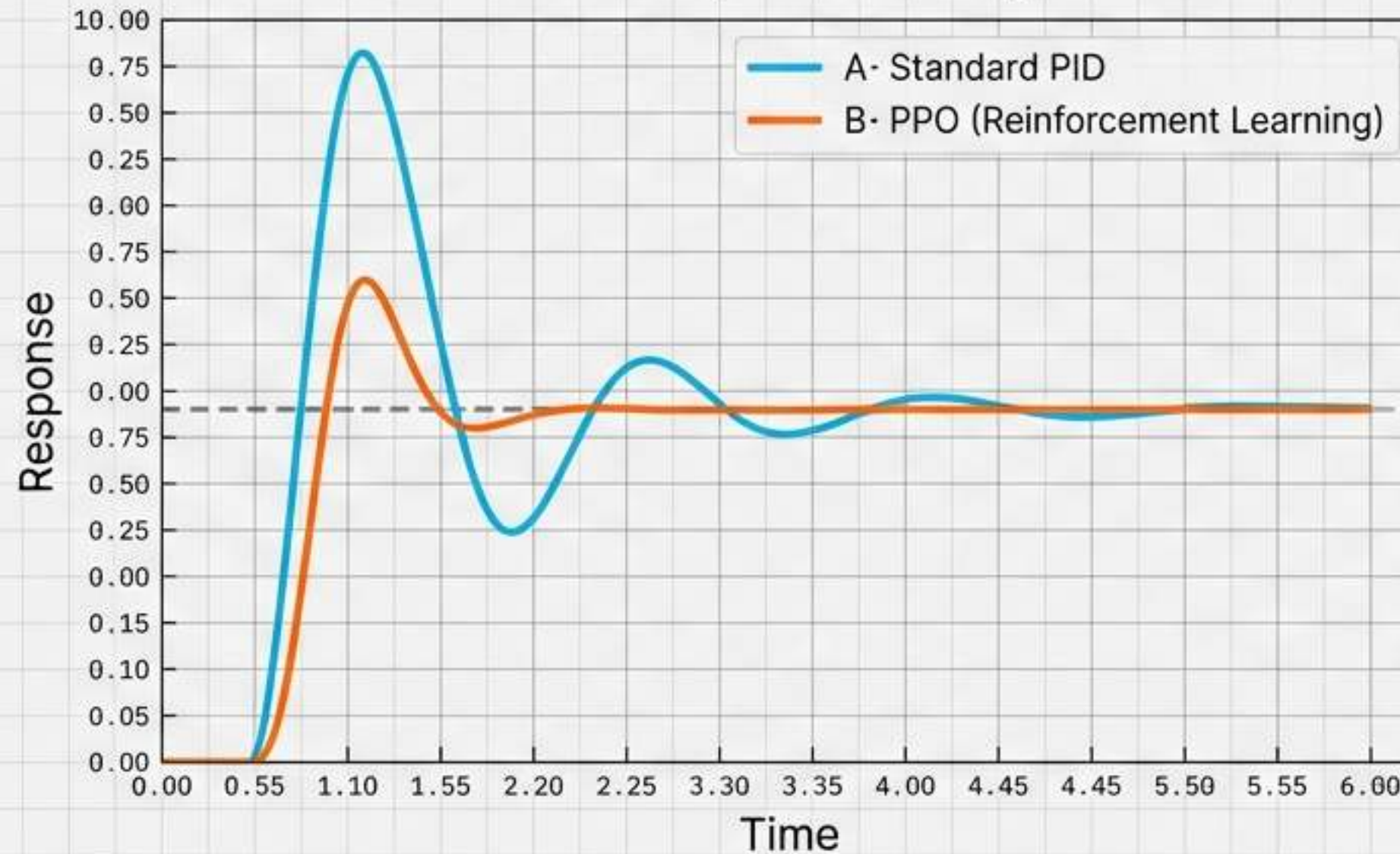
The Brain II: The PID Loop



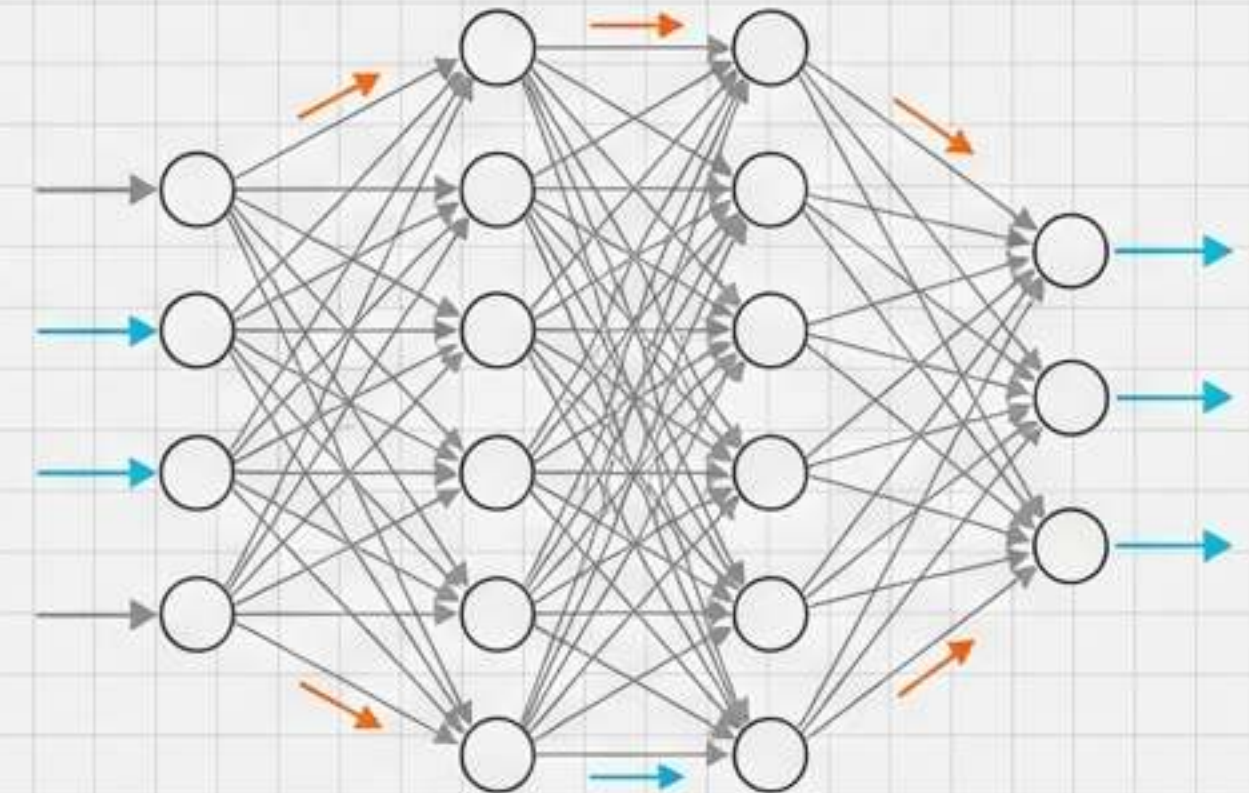
This mathematical heartbeat runs 4,000 times per second.

The Brain II: Evolution

Controller Response Comparison

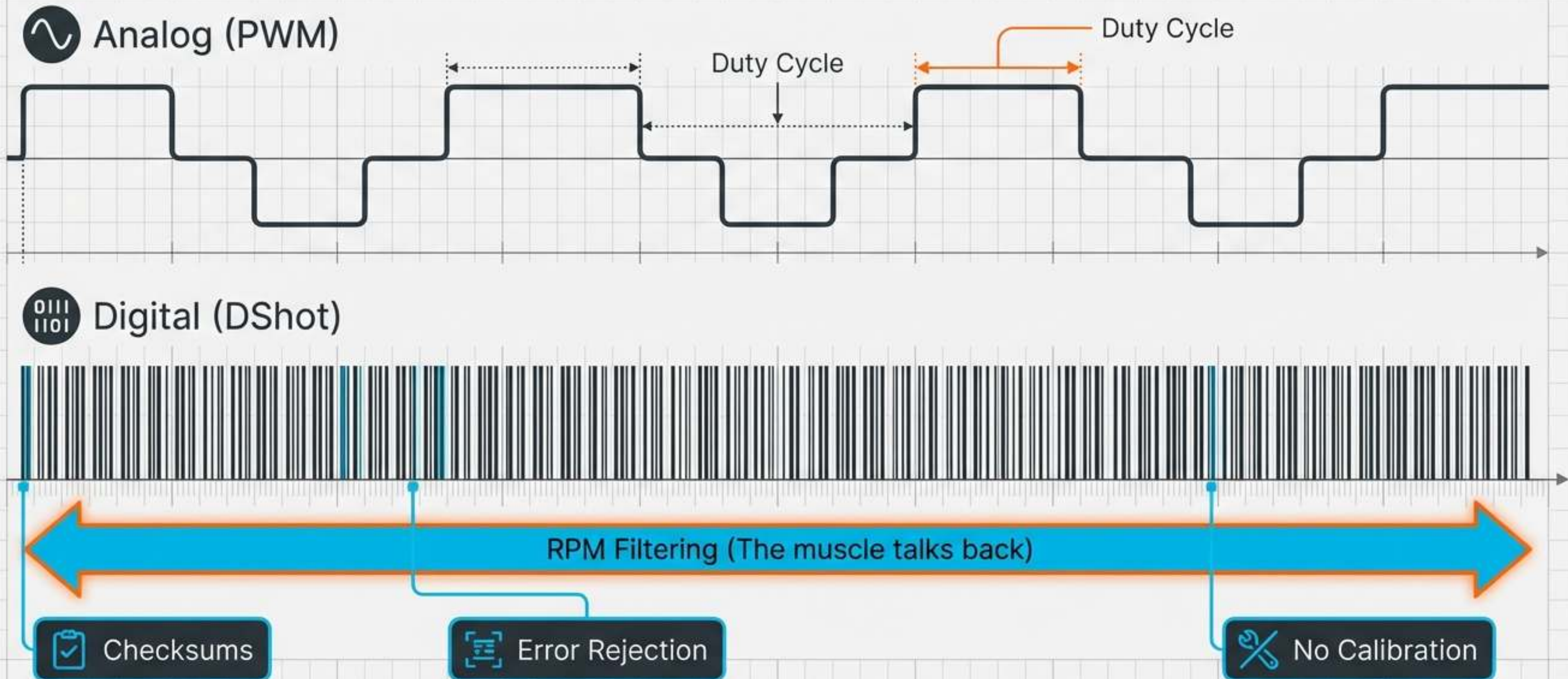


Neural Network Controller

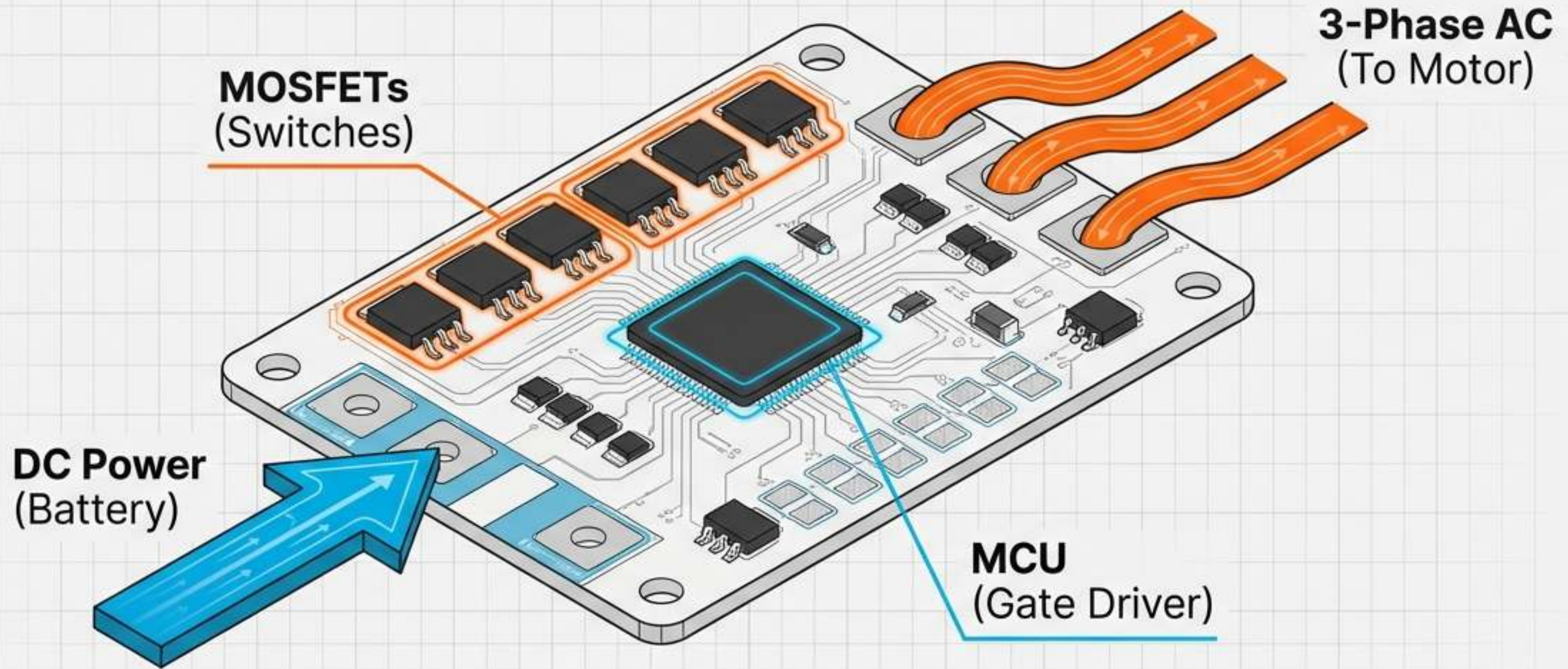


End-to-End Learning. The drone learns physics in a simulator (Digital Twin) before ever flying.

The Translator: Protocols.

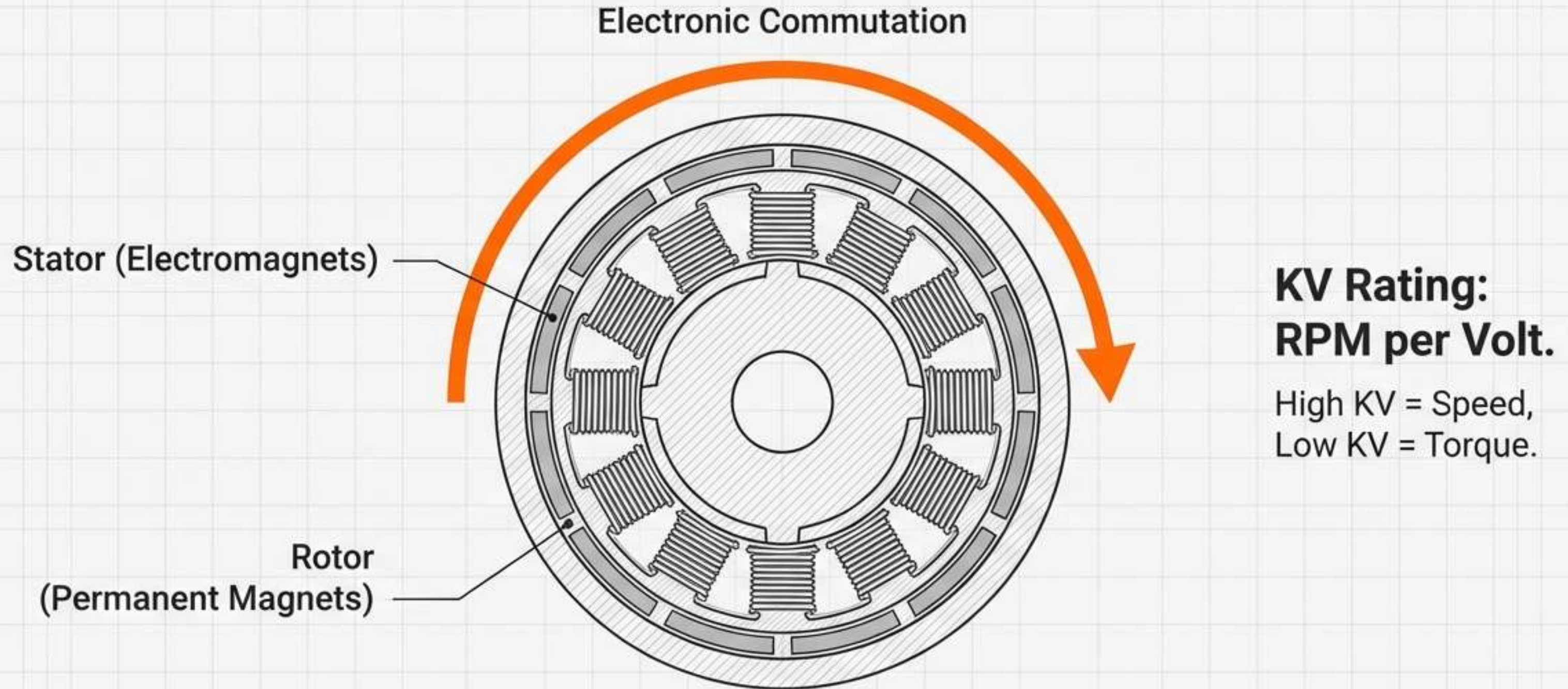


The Muscle I: ESC

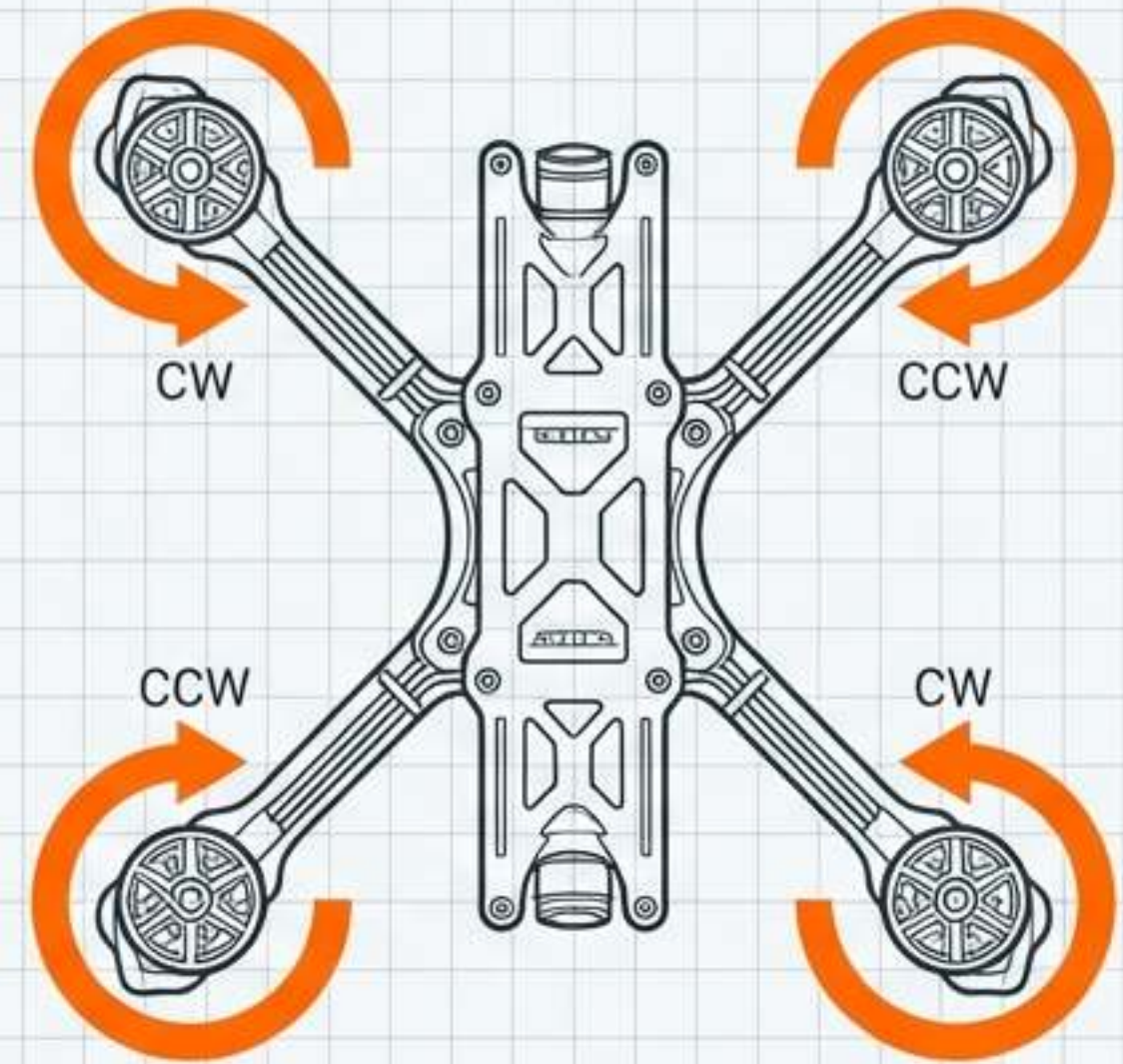
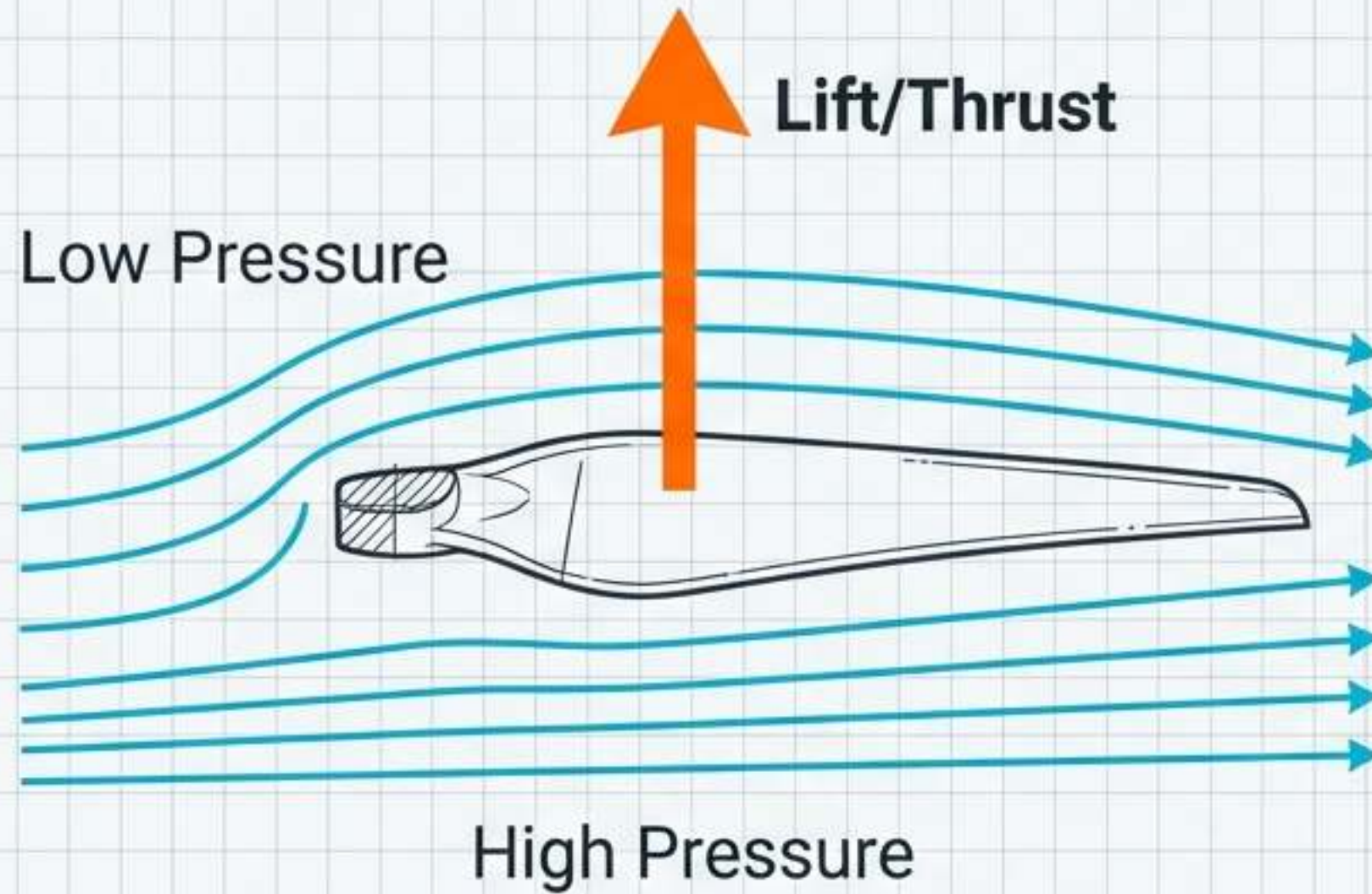


Switching power 50,000 times a second to create a rotating magnetic field.

The Muscle II: Brushless Motors.



Interaction: Aerodynamics

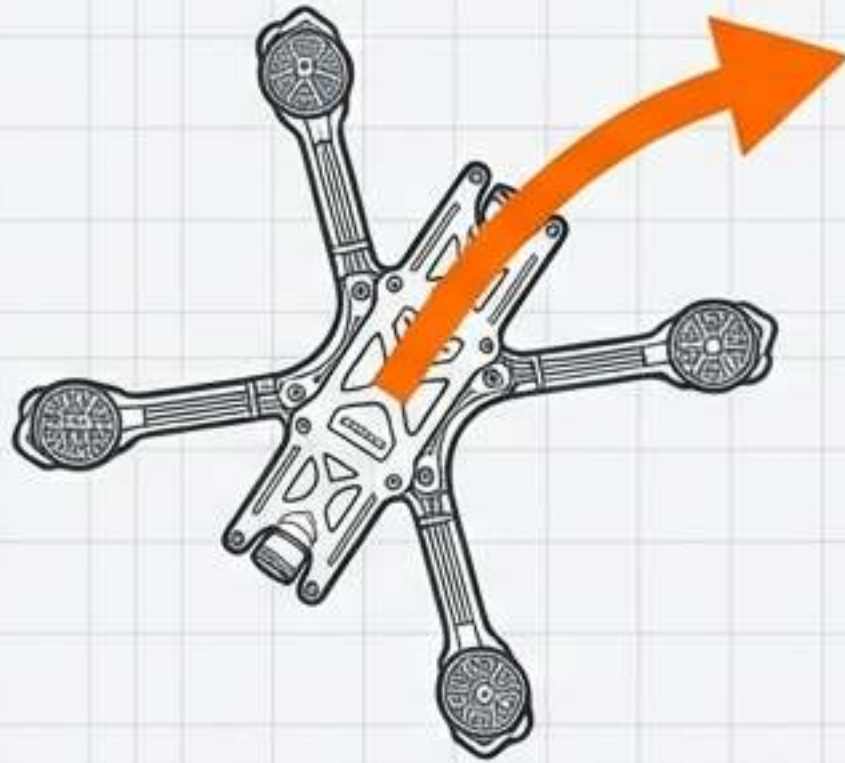


Torque Cancellation.
Opposing spins neutralize Yaw.

Unbalanced props create vibration, the enemy of the IMU

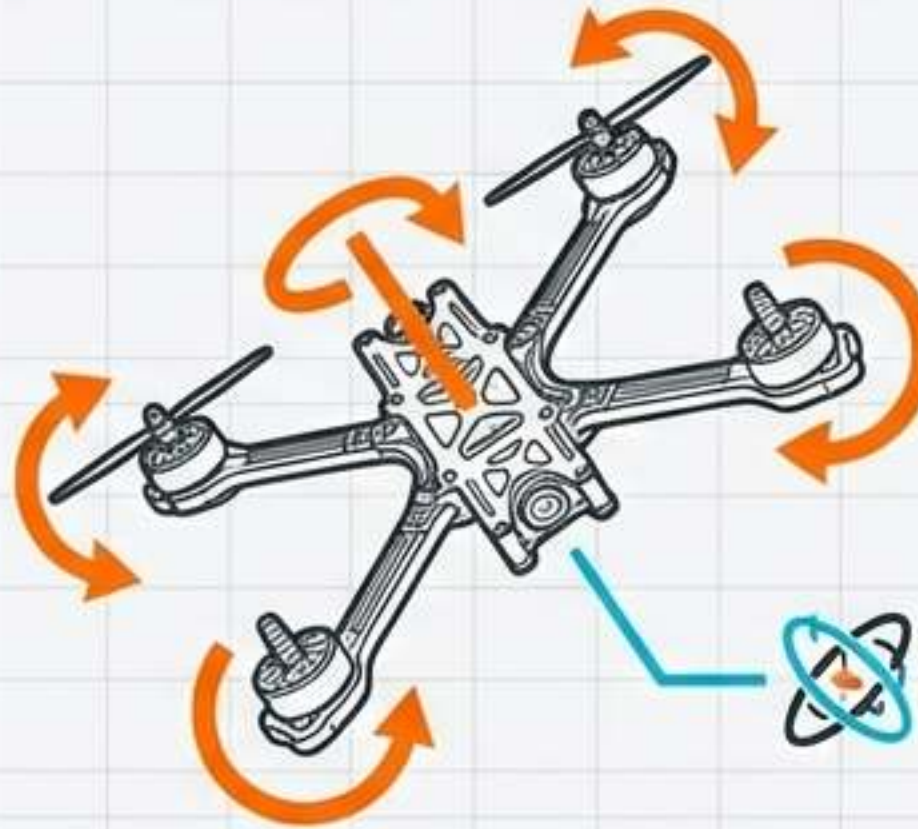
The Result: Flight Modes.

Angle Mode (Self-Level).



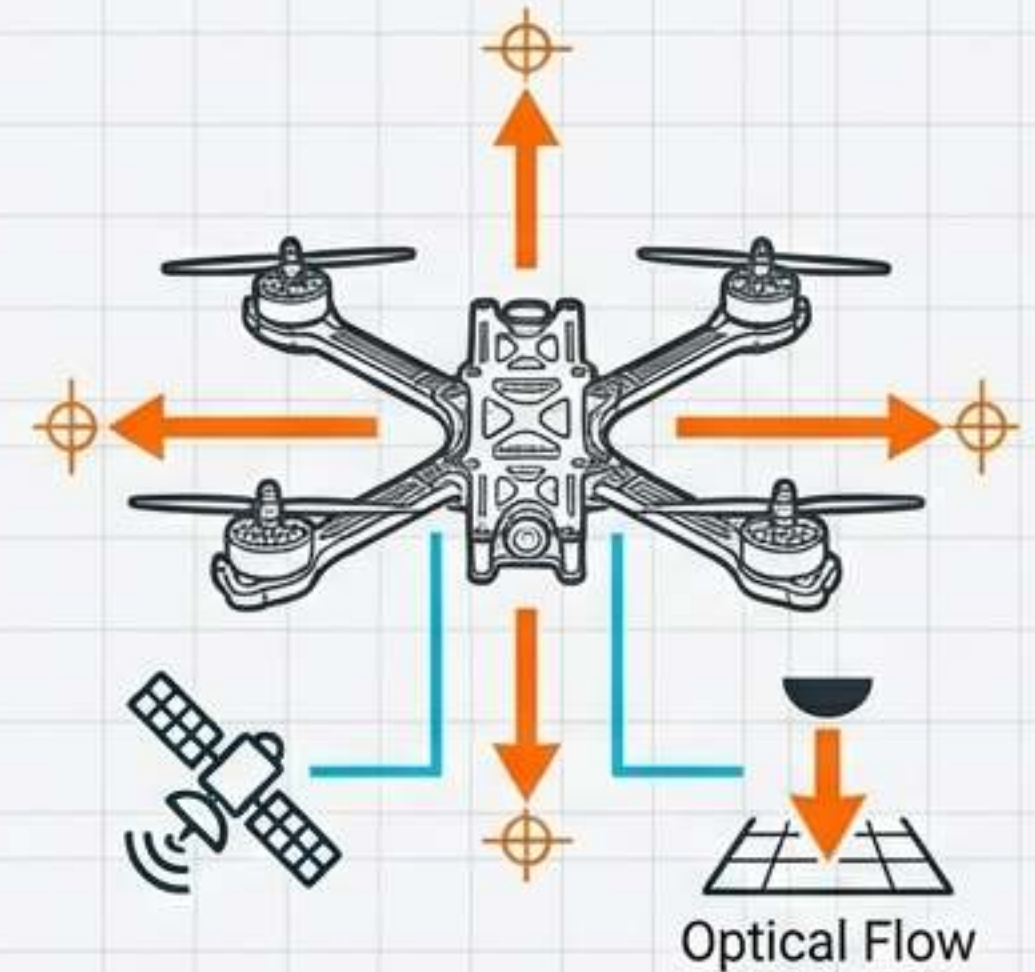
Stick = Desired Angle.
Accelerometer Dominant.

Acro Mode (Rate).



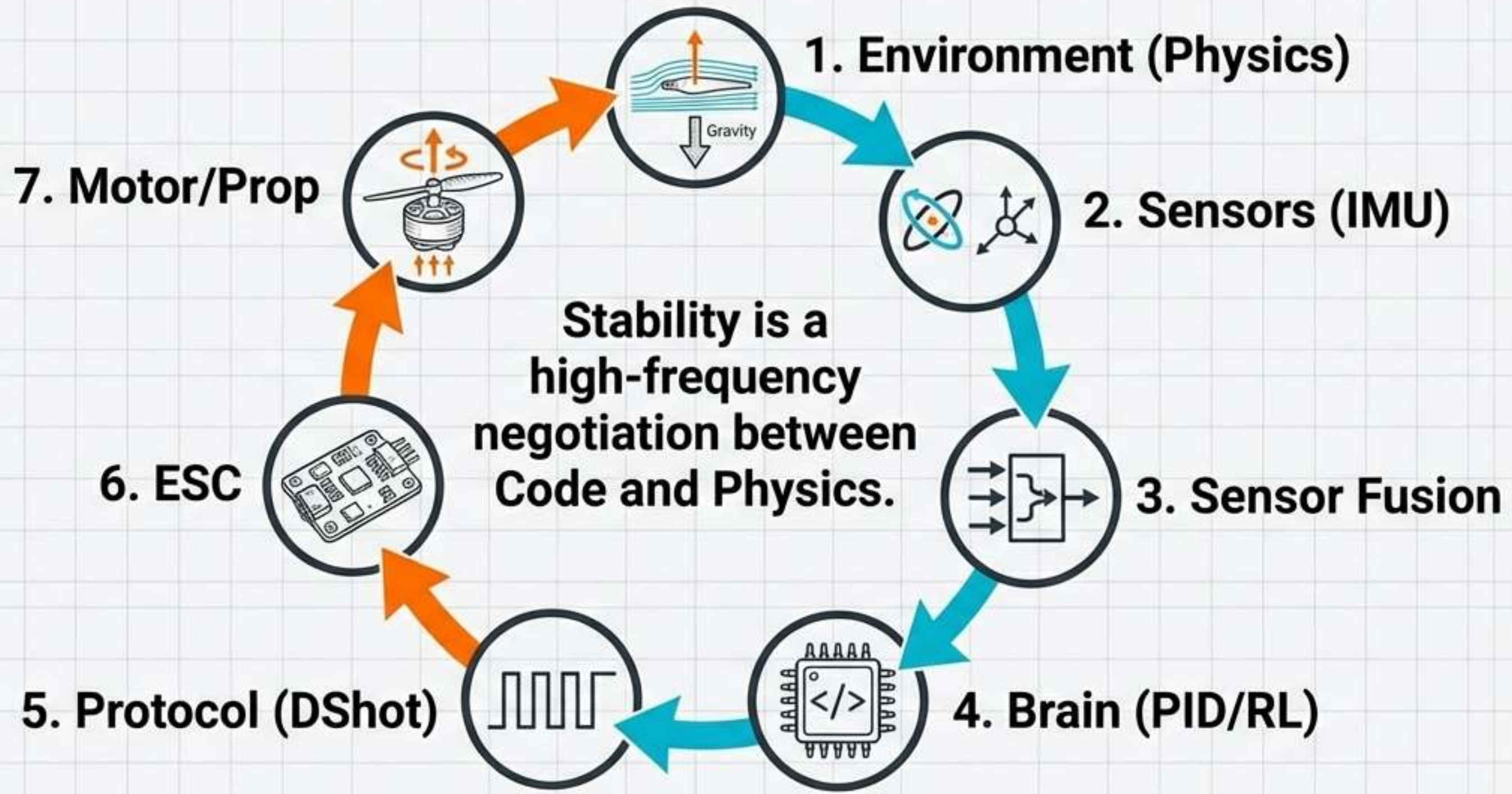
Stick = Rotation Speed.
Gyroscope Dominant.
No Limits.

Position Hold.



Stick = Velocity.
GPS + Optical Flow.

Summary: The Digital Organism.



Future Outlook: Adaptive AI agents and bi-directional digital feedback.